

SMART VACUUM CLEANER AGENT

NAME

SAAD NAWAZ

SEMESTER

BSCS – 6th

SECTION / ROLL
NO.

‘C’ - 23122157

AGENT OVERVIEW

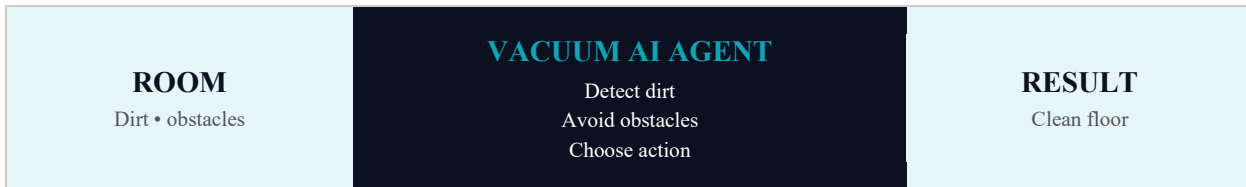
SMART VACUUM CLEANER

What does the agent do?

The Smart Vacuum Cleaner is an AI-based reflex agent that cleans a room automatically. It senses dirt and obstacles on the floor, decides whether to clean, move, or turn, and acts through its motors and brushes. When the battery is low, it returns to its charging dock.

SENSE → DECIDE → ACT

Graphical Representation



Environment

ROOM + FLOOR + FURNITURE + DIRT + CHARGING DOCK

PEAS

PEAS Framework

P	Performance Measure Cleanliness achieved, time and energy used, area covered, avoiding damage to furniture, and battery management.
E	Environment Room, floor type, furniture, dirt, obstacles, and charging dock location.
A	Actuators Drive wheels, brushes, suction motor, and status lights.
S	Sensors Dirt sensor, bump/obstacle sensor, cliff sensor, and battery-level sensor.

ENVIRONMENT TYPE

Agent Type

Simple Reflex Agent (with a basic model of dirt and obstacles)

Observability	Partially Observable	The agent only senses its immediate surroundings, not the whole room layout.
Determinism	Stochastic	Real-world dirt, obstacles, and slipping wheels make outcomes unpredictable.
Episodes	Sequential	Each move affects the agent's position and what it senses next.
Change	Dynamic	People, pets, and furniture can move while the agent is cleaning.
Values	Discrete	Dirty/clean, obstacle/no obstacle, and battery level are treated as discrete states.
Agents	Single Agent	One vacuum agent operates and cleans the room independently.